Quiz 2

ETL — the pipeline, and the numbers it must hit

Covers	Power Query, live; and reading a wrong total back to its step
Tasks	T4, T5, T6 and T7 of the ETL workbook, **and the hands-on lab**
Marks	**51** (of the 95 workbook marks for this dataset)
CLO	CLO3 · Courage
Component	Hands-on Exercises (20% of the course)
Work	**Individual**
Tool needed	**Power BI Desktop, Power Query in Excel, or pandas**
Submit	A **GitHub repository link** on the LMS — folder `quiz2/`
Due	As posted on the LMS

What this quiz is

Quiz 1 was the paper design: the defect census, stage ownership, and the order of the seventeen steps. **This quiz is the pipeline, and the numbers it has to arrive at.**

T4 to T7 cannot be answered from the slides. Every figure comes out of `data/source`, and T7 will not pass until you have actually built the thing. Build it first, then fill the sheets.

> **Do Quiz 1 before this one.** T4 reads the gap between two totals and names the step that caused it — which assumes you have a step list. T6 asks which of the eight checks catches each failure. Both are Quiz 1 arriving as numbers.

The deliverable

A one-off merge produces a table. **This quiz produces the thing that produces the table.**

```
THREE SOURCES              STAGING                  THE STAR
(read-only)                (never user-facing)      (everybody)

marine .csv  --+        E1–E3  land, log           DimDate   28
road   .csv  --+------> T1–T7  per source   ------> DimCity   15
rail   .xlsx --+        T8–T10 once, combined      DimMode    4
                                                   DimStatus  4
37 rows                 35 appended                DimRoute  21
10 defects              34 after dedup             FactShipment 34
```

`FactShipment.csv` is deliberately **not** in the data pack. It is the finish line, not the starting point. Producing it is the lab.

Shape, then text, then type, then keys. Four words in that order will get you through most of the lab and most of the midterm practical.

The data you need

Everything is in the folder distributed with this quiz.

File	What it is
`data/PakFreight_ETLLab_STUDENT.xlsx`	Sheets T4, T5, T6 and T7 are this quiz
`data/source/*`	**The three dirty files, unmodified.** 37 rows, ten defects. The input
`data/source/night2/*`	**The refresh test.** Same marine and rail; a road file your pipeline has never seen
`data/dims/*.csv`	The five dimensions of the star. **Import them; do not rebuild them**
`data/alias_lookup.csv`	The 23-row alias map — to **check** your own against, not to import
`PowerQuery_Lab_Guide.md`	The walkthrough: nine steps, the M code for all three queries, the refresh test, troubleshooting — **plus a Python appendix** with the whole pipeline in pandas

> **Do not open `night2/` until T7 part A is finished and all eight checks pass.** It is there to test a pipeline, not to help you build one.

Power Query or Python — your choice, same marks. Every number you have to hit is identical, because the defects are facts about the data and not about the tool. The four things required in place of named applied steps are listed under *The Python track* below.

The rules

- **A pipeline that reconciles is not the same as a pipeline that is right.** 578,650 kg and PKR 4,808,000 are the weakest checks you have — a locale bug and a status casing bug both pass them to the rupee.
- **Nothing is dropped silently.** A row you cannot resolve goes to the Unknown member at `-1` and is **counted**. Deleting it is an answer, and you have to defend it in writing.
- **The totals must not move.** Cleaning data never changes what the business shipped.
- **Name every step.** It is the only documentation the pipeline will ever have, and it is marked.
- **T7 marks itself.** Every `fix` means something earlier is still wrong. Go and find it; do not adjust the check.

T4 · Row arithmetic · 15 marks

Every wrong total here is 578,650 plus or minus something specific, and the something tells you which step failed. Learn to read the gap.

Part A · The ladder · 6 marks

Fill it in **from the files**, not from the slides.

Stage	Rows	SUM(WeightKG)	SUM(FreightCostPKR)
marine, as extracted	?	?	?
rail, as extracted	?	?	?
road, as extracted	?	?	?
raw rows landed in staging	?	—	—
appended	?	?	?
loaded into `FactShipment`	?	**578,650**	**4,808,000**

> The road row is the one that catches people: **fifteen rows land and thirteen survive**, and the road subtotal still has the duplicate inside it. That is why the road rung and the loaded rung differ by 6,100 kg and not by nothing.

Part B · Read the gap · 9 marks

Three real totals a student has handed in. Work out the arithmetic that produced each, then name the step that was missing or misplaced:

```
SUM(WeightKG) = 283,695.25 kg
SUM(WeightKG) = 584,750 kg
SUM(WeightKG) = 661,650 kg
```

Then two more symptoms that move **no total at all** — headers reading `Column1, Column2, ...`, and a fact table of 35 rows whose totals are both exactly right. Name the cause of each.

A row count is not a total, and a total is not a row count. That second symptom is the entire reason check 5 exists separately from checks 7 and 8.

T5 · The locale trap · 12 marks

The marine file writes its dates as text in `dd/MM/yyyy`. On a machine set to United States regional settings, Power Query reads `01/02/2026` as the **second of January**.

Fill in all twelve strings read **both ways**, then classify each row as **Right by luck**, **Wrong month** or **Error**.

Self-check (in the sheet): 10 wrong, **1** right by accident, **1** error — and the eleven readable rows land in **eleven different months**.

August never appears, because nobody shipped on the 8th. If you can say why that matters, you have understood that "check the month list" and "check the range" are not the same check.

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6 of the 12 marks are for three written answers:

1. Which of the three files is exposed to this, and how many loaded rows that puts at risk. (The other two are not exposed, and for two different reasons.)
2. Both reconciliation totals are exact and every monthly report is still wrong. Explain why the totals **cannot** catch it.
3. The single click that prevents it, and the check that catches it if you forget.

This is the only defect of the ten that produces a plausible, confidently wrong answer with **no error message anywhere**. "The dates were wrong" earns nothing without naming the click.

T6 · Loud or quiet · 12 marks

Six failures — four scheduling accidents and two defects. For each: loud or quiet, which of the eight checks would catch it, and what makes it safe.

A **loud** failure stops the run: a job goes red, somebody is paged, the warehouse is unchanged, and the 08:00 report shows yesterday's number with a stale-as-of stamp on it. Embarrassing, bounded, fixable before anyone trades on it.

A **quiet** failure succeeds. Eleven rows went to Unknown, one region is missing, the dates landed in the wrong month, and the dashboard is green. Nobody finds out until a quarter closes.

Self-check (in the sheet): only **1** of the six is loud — and it is the one you have no work to do about.

4 of the 12 marks are for the answer that is not the obvious one. Everybody says they would rather have the loud failure. State what follows from that about **where the engineering effort in a pipeline actually goes**.

A pipeline that never fails is not reliable. It is unmonitored.

T7 · Definition of done · 12 marks

Refresh your model, read each number off it, and type it in. **All eight must read "pass".** Submitting with a single `fix` loses all 12 marks.

#	Check	Expected
1	Contract columns in each of the three queries	**7**
2	Rail subtotal, in kilograms	**295,250**
3	Distinct values in `StatusRaw`	**3**
4	Latest day in `ShipDate`	**13** February 2026
5	Rows in `FactShipment`	**34**
6	Rows carrying a `-1` city key	**0**
7	`SUM(WeightKG)`	**578,650 kg**
8	`SUM(FreightCostPKR)`	**PKR 4,808,000**

Checks 1 to 4 come first, and that is not an ordering accident. Checks 7 and 8 are the famous ones and they are the weakest on the sheet. Work them in order — if 7 fails, fix 2 and 5 first, because they cause it.

Check 6 is the character question

Check 6 expects **0**, and that is the trap. Zero is right here only because the alias map happens to cover all 23 spellings these three files contain. It is not a law of nature, and **"0" typed without looking is the same answer as "0" typed after looking.**

So the sheet asks *how you know* — and then asks what the same pipeline reports on `night2`.

> Integrity is logging the rows you could not fix. Routing a row to Unknown is not forgiveness; dropping it quietly so the dashboard looks complete is a statement about you, not about the data.

Part B · The refresh test

Point your three queries at `data/source/night2/` and press **Refresh**. Nothing should need fixing. Then fill in six more numbers.

That folder holds the same marine and rail files and a road file you have never seen: an extra consignment, a second copy of `RD-508` that differs from the first by **one trailing space**, a status typed as `DELAYED`, and a shipment from a city no alias resolves.

	Expected
Rows in `FactShipment`	**35**
`SUM(WeightKG)` / `SUM(FreightCostPKR)`	**584,950 kg** / **PKR 4,860,000**
Rows carrying a `-1` key	**1** — and you name the city
Rows with status `Delayed`	**5**
If you deduplicated before trimming	*36 rows, 591,550 kg*

> If you get **36 rows and 591,550 kg**, you deduplicated before you trimmed. That is the Quiz 1 constraint arriving as a number, and nothing in the first night's file would ever have told you.

The lab · Rebuild the merge as a pipeline

Full walkthrough in `PowerQuery_Lab_Guide.md` — nine steps, the M code for all three queries, where every click lives, and a troubleshooting table that reads a wrong total back to the step that caused it.

In outline:

Step	What you do
1	Shape the rail workbook first: skip two title rows, **then** promote headers
2	Drop the road file's blank line and `TOTAL` footer — two named steps, not one
3	Rename all three into the seven-name contract
4	Trim, clean, Proper Case, uppercase, `RL–` prefix; dates **Using Locale**; tons x 1,000; `ModeKey` by source
5	Append the three, then deduplicate on `ShipmentID` — 35 in, 34 out
6	Split `RawAliases` into rows, merge **left outer** twice, expand, nulls to `-1`
7	Staging queries to connection-only, refresh, clear all eight checks
8	**The refresh test** against `night2`
9	If you finish early: `DateKey`, `StatusKey`, `RouteKey`, and build `DimRoute` yourself

How many applied steps you should have

Query	Steps	Why it differs
`Road`	**11**	the reference query
`Marine`	**9**	no blank row and no `TOTAL` footer, so two filters do not exist
`Rail`	**13**	adds navigate-to-sheet, skip title rows, the `RL–` prefix, the tons conversion
`FactStaging`	**5**	append, dedup, merge, expand, replace nulls

> If your marine query has eleven steps you have added two that do nothing, and a filter that removes nothing is **worse** than no filter: the next person reads it as evidence that the file has junk rows.

Why the refresh is the point

Replace the three source files with next month's three, click **Refresh**, and all sixteen steps replay: headers promoted, junk dropped, names conformed, tons converted, duplicates removed, keys assigned, Unknown routed, star loaded.

A hand-built merge produces a table. Today you produce **the thing that produces the table** — and if anything needs fixing by hand, your pipeline is still a one-off.

The Python track

You may do this lab in pandas instead of Power Query, and you are marked the same. `Power-Query_Lab_Guide.md` ends with an appendix carrying the whole pipeline, ready to run. `pip install pandas openpyxl`.

Nothing about the task changes. The ten defects, the four transform classes, the three ordering constraints and all eight checks are facts about **the data**, not about the tool.

The traps are all still there:

Trap	In pandas
The locale	`pd.to_datetime("01/02/2026")` gives **2026-01-02**. pandas guesses month-first and makes the exact mistake. Naive parsing spreads these 12 rows over **eleven months**
Append before rename	`pd.concat` of three un-renamed frames gives **20 columns**, only `Status` shared, **no error**
Dedup before trim	night two loads **36 rows / 591,550 kg** instead of 35 / 584,950
Tons before append	**283,695.25 kg**, short by 294,954.75

What you must do instead of naming applied steps

There is no applied-steps pane, so four things are required in its place. Each one loses the same mark its Power Query equivalent would:

	Required	Instead of
1	A named `STEPS` list, keeping each intermediate frame so you can look at step 7	renaming applied steps
2	`format="%d/%m/%Y"` stated explicitly — a bare `pd.to_datetime` loses the mark	Change Type **Using Locale**
3	`drop_duplicates(subset=["ShipmentID"])` — never bare	the Remove Duplicates dialog
4	`how="left"` then `fillna(-1)` — `how="inner"` drops the row and moves the total	Merge Queries, Left Outer

Plus one that replaces **"Enable load, off"**: say which frames are intermediate and which are published. No setting marks the staging boundary for you, so you mark it yourself.

`dayfirst=True` is not the same as stating a format. It is a hint pandas may override. State the format.

> **Before you choose.** The **midterm practical** is an ingestion pipeline and a star, **under time**, and the sessions after it are Power BI visualization and DAX. A pandas pipeline earns full marks here — and leaves you meeting Power Query cold in the exam. If you are unsure: do the workbook in either, and the lab in Power Query.

Marking guide

Task	What is marked	Marks
T4	The ladder, filled from the files (6) + three wrong totals read back to their step, and the two symptoms that move no total (9)	**15**
T5	Twelve strings read both ways and classified (6) + the three written answers, one of which must name the click (6)	**12**
T6	Six failures, loud or quiet, with the check that catches each (8) + where the engineering effort actually goes (4)	**12**
T7	All eight checks reading **pass** (8) + check 6 answered with *how you know*, and the six refresh-test numbers (4)	**12**
	Quiz 2 total	**51**
Lab	The pipeline itself: three source queries with the step counts above, every step named, staging connection-only, 34 fact rows, both totals exact, the refresh test run	*gate*

The lab is a gate, not a score. A workbook submitted without a pipeline behind it cannot have passed T7 honestly, and T7's twelve marks go with it.

A single `fix` cell in T7 loses all 12 marks for T7. That is not a punishment; it is the definition of done.

How to submit

You submit a link, not a file. Same repository as Quiz 1, new folder.

1. Use the repository you made for Quiz 1:

```
DW-<your roll number>
```

2. Create this folder inside it, exactly:

```
quiz2/
```

3. Put these files in that folder:

File	What it is
`Quiz2_<roll number>.xlsx`	The workbook with **all seven sheets** filled. T4–T7 are what is marked here; T7 must read **8 / 8** and **6 / 6** on the refresh test
`pipeline.pbix` — **or** `pipeline.py`	Power Query: three source queries with named applied steps, staging connection-only, six tables in the model. Python: the named `STEPS` list, the eight checks as assertions, the `-1` count printed
`FactShipment.csv`	Your output. **34 rows**, 578,650 kg, PKR 4,808,000
`refresh_test/`	The before-and-after screenshots of the `night2` refresh
`answers.md`	The written parts: what makes two rows a duplicate and why that test cannot run before the trim; what happens to a row whose city you cannot resolve, and why that is not "delete it"; the `-1` count **as a number** for both nights; the locale click; where the engineering effort goes
`README.md`	Your name, roll number, and which tool you used — Power BI Desktop, Power Query in Excel, or pandas

4. Commit and push:

```
git add quiz2
git commit -m "Quiz 2"
git push
```

5. On the LMS, submit the repository link only:

```
https://github.com/<your-username>/DW-<your roll number>
```

> **Commit the `.pbix` to your own repository** — it is the deliverable here, so it belongs in the submission. If it is over 50 MB GitHub will warn you; a file of this size will not be.
>
> If your repository is private the submission cannot be marked. Check your own link in a logged-out browser window before you submit it.

Definition of done

- ☐ Three source queries, step counts **11 / 9 / 13**, every step **named**
- ☐ `Marine`, `Rail`, `Road`, `AliasLookup` all connection-only
- ☐ Six tables in the model; ten queries behind it
- ☐ `AliasLookup` has **23** rows and **14** distinct keys, not 24
- ☐ Every date typed **Using Locale**, English (United Kingdom) — never plain Change Type
- ☐ Rail subtotal **295,250 kg** *before* the append
- ☐ All eight checks pass — **8 / 8**
- ☐ The `-1` count written down **as a number**, for both nights
- ☐ The refresh test run against `night2`, with a before-and-after screenshot, reading **35 rows / 584,950 kg / 1 unknown / 5 delayed**
- ☐ **578,650 kg** and **PKR 4,808,000**, exactly
- ☐ `FactShipment.csv` committed, 34 rows
- ☐ `answers.md` and `README.md` present
- ☐ Pushed, public, and the link opens for a stranger

References

- **Kimball & Ross (2013)**, *The Data Warehouse Toolkit*, 3rd ed. — Ch. 19 on the ETL subsystems; Ch. 20 on staging, dependency order and audit columns; Ch. 3 §"Dimension Table Attributes" for conformed values and the Unknown member from first principles.
- **Kimball & Caserta (2004)**, *The Data Warehouse ETL Toolkit* — Chapters 4 and 5 on cleaning, conforming and error-event handling.
- **Inmon (2005)**, *Building the Data Warehouse*, 4th ed.
- **Microsoft Learn** — "Power Query M formula language reference", "Query folding guidance", "Shape and combine data in Power BI Desktop".
- **Kimball Group** — "The 34 Subsystems of ETL".

One sentence to leave with. Cleaning a file is one job and building the thing that cleans it is another, so the deliverable was never the table: it is the seventeen steps, in order, with their names on them — because those are what still work tomorrow night at two in the morning when nobody is watching.
